

Bonus task 2

14.9.2025

Local and global stiffness matrices, boundary conditions, and loadings

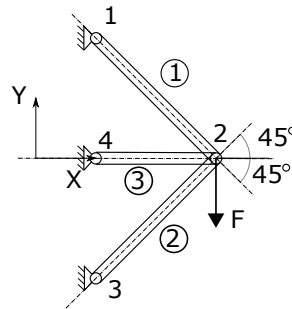
Possibility to earn five bonus points in total:

- **2 points** from the submitted MATLAB/Python assignment and report.
- **3 points** from short pen-and-paper quizzes during the lecture the 28th of September.

Deadline for the submitted assignment: Before the lecture on the 28th of September.

The aim of the submitted assignment is to solve the global nodal displacements u_2 and v_2 at node 2 of the simple truss structure shown in the Figure. After solving the nodal displacements, determine the axial member forces for Trusses 1, 2, and 3.

Use the node and element numbering shown in the Figure.



The parameters of the problem are given in Table 1.

The cross-sectional area of truss 3 depends on the last digit of your student number. Let n_{student} denote the last digit of your student number, where $n_{\text{student}} \in \{0, 1, \dots, 9\}$. The cross-sectional area of Truss 3 is

$$A_3 = 10 + 4n_{\text{student}} \quad [\text{mm}^2]. \quad (1)$$

For example, if the last digit of your student number n_{student} is 7,

$$A_3 = 10 + 4(7) = 38 \text{ mm}^2. \quad (2)$$

Use the value of A_3 corresponding to your own student number throughout the calculations and FE analysis.

Table 1. Material and geometric parameters of the structure.

Member	Young's modulus, E [MPa]	Cross-sectional area, A [mm ²]	Length, L [mm]
Truss 1	210000	100	1000
Truss 2	210000	50	1000
Truss 3	210000	$10 + 4n_{\text{student}}$	$1000 \cos(\pi/4)$
Applied force, F [N]	100000		

Write a short report where you give answers for the following:

Submission and bonus points

This assignment gives a maximum of **2 bonus points**. Additional bonus points related to this topic will be available through short pen-and-paper quizzes during the lectures.

Write a short report containing the following results:

1. Calculate the global nodal displacements u_2 and v_2 at node 2 using the value of A_3 corresponding to your student number. Present the results with appropriate units. (1 point)
2. Calculate the axial member forces. State whether each member is in tension or compression. (1 point)

Present the numerical results clearly in tables.

Submit in Moodle:

- A short PDF report.
- All MATLAB (.m) or Python (.py) files required to reproduce the results.

The submitted code must run without additional editing. Both the PDF report and runnable source code are required to receive points.

Present all numerical results in tables (include units where applicable) and submit in Moodle:

- A PDF report.
- All MATLAB (.m) or Python (.py) source files required to reproduce your results.

Make sure that the code runs without extra editing (provide a main script and functions).

Note: No points will be awarded without both the PDF report and runnable source code.

Please note that code templates KlocRod.m, TRod.m, and run_linStaticRod.m for MATLAB and KlocRod.py, TRod.py, and run_linStaticRod.py for Python are provided.

Tips

In this task, you can use a ready solution for a stiffness matrix. As you know from lectures and literature (see for example, Cook p.20 Eq. (2.2.1) or Hakala p.54 Eq. (2.3.17)), the stiffness matrix of a rod element can be written as follows:

$$\overline{\mathbf{K}} = \frac{EA}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad (3)$$

Analyzed structures are not often parallel to the element's longitudinal direction -> local elemental stiffnesses need to be present in the global coordinate system.

$$\mathbf{K}_G = \mathbf{T}^T \overline{\mathbf{K}} \mathbf{T} \quad (4)$$

where $\mathbf{K}_G$ is a global elemental stiffness matrix (of size 4×4 for a two-nodded rod element) and $\mathbf{T}$ is a transformation matrix that can be written as follows:

$$\mathbf{T} = \begin{pmatrix} \cos \alpha & \sin \alpha & 0 & 0 \\ 0 & 0 & \cos \alpha & \sin \alpha \end{pmatrix} \quad (5)$$

Transformation needs to be done for all (two) elements.

$$\mathbf{K}_G^{(1)} = \mathbf{T}^T(\alpha_1) \bar{\mathbf{K}}^{(1)}(E_1, A_1, L_1) \mathbf{T}(\alpha_1) \quad (6)$$

$$\mathbf{K}_G^{(2)} = \mathbf{T}^T(\alpha_2) \bar{\mathbf{K}}^{(2)}(E_2, A_2, L_2) \mathbf{T}(\alpha_2) \quad (7)$$

$$\mathbf{K}_G^{(3)} = \mathbf{T}^T(\alpha_3) \bar{\mathbf{K}}^{(3)}(E_3, A_3, L_3) \mathbf{T}(\alpha_3) \quad (8)$$

and global elemental stiffness matrices look like that

$$\mathbf{K}_G^{(1)} = \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} & k_{13}^{(1)} & k_{14}^{(1)} \\ k_{21}^{(1)} & k_{22}^{(1)} & k_{23}^{(1)} & k_{24}^{(1)} \\ k_{31}^{(1)} & k_{32}^{(1)} & k_{33}^{(1)} & k_{34}^{(1)} \\ k_{41}^{(1)} & k_{42}^{(1)} & k_{43}^{(1)} & k_{44}^{(1)} \end{pmatrix} \begin{matrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{matrix} \quad (9)$$

$$\mathbf{K}_G^{(2)} = \begin{pmatrix} k_{11}^{(2)} & k_{12}^{(2)} & k_{13}^{(2)} & k_{14}^{(2)} \\ k_{21}^{(2)} & k_{22}^{(2)} & k_{23}^{(2)} & k_{24}^{(2)} \\ k_{31}^{(2)} & k_{32}^{(2)} & k_{33}^{(2)} & k_{34}^{(2)} \\ k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} \end{pmatrix} \begin{matrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix} \quad (10)$$

$$\mathbf{K}_G^{(3)} = \begin{pmatrix} k_{11}^{(3)} & k_{12}^{(3)} & k_{13}^{(3)} & k_{14}^{(3)} \\ k_{21}^{(3)} & k_{22}^{(3)} & k_{23}^{(3)} & k_{24}^{(3)} \\ k_{31}^{(3)} & k_{32}^{(3)} & k_{33}^{(3)} & k_{34}^{(3)} \\ k_{41}^{(3)} & k_{42}^{(3)} & k_{43}^{(3)} & k_{44}^{(3)} \end{pmatrix} \begin{matrix} u_4 \\ v_4 \\ u_2 \\ v_2 \end{matrix} \quad (11)$$

Assembling

Sometimes, it is helpful to present elements' connectivity and corresponding (global) nodal displacements in the table as follows:

	Node I	Node J	nodal displacements
Truss 1	1	2	u_1, v_1, u_2, v_2
Truss 2	2	3	u_2, v_2, u_3, v_3
Truss 3	4	2	u_4, v_4, u_2, v_2

Note that $u_1, v_1, u_2, v_2, u_3, v_3, u_4, v_4$ are the structure's global displacement coordinates. Common stiffness components (interconnected via the same node) of global elemental stiffness matrices $\mathbf{K}^{(1)}$, $\mathbf{K}^{(2)}$ and $\mathbf{K}^{(3)}$ can be seen from the connectivity table, and they are needed to sum to mimic structure correctly. After assembling, the global stiffness matrix should be a size of 8×8 .

Boundary conditions and linear solution

The rods are fixed to the wall via hinged supports (at nodes 1 and 3). Therefore, global nodal displacements $u_1 = 0, v_1 = 0, u_3 = 0, v_3 = 0, u_4 = 0, v_4 = 0$ are restricted to displace. After eliminating corresponding rows and columns of the global stiffness matrix and the external force vector, global displacements can be solved.

$$\mathbf{u}_{G_{red}} = \mathbf{K}_{G_{red}}^{-1} \mathbf{f}_{G_{red}} \quad (12)$$

Member forces

After solving $\mathbf{u}$, local elemental displacements for an element i can be compute as follows:

$$\bar{\mathbf{u}}^{(i)} = \mathbf{T} \mathbf{u}_G^{(i)} \quad (13)$$

and based on this, member forces for an element i can be found as

$$\bar{\mathbf{f}}^{(i)} = \bar{\mathbf{K}}^{(i)} \bar{\mathbf{u}}^{(i)} \quad (14)$$

So, displacements at the local elemental level for rod 1:

$$\bar{\mathbf{u}}^{(1)} = \mathbf{T}(\alpha_1) \mathbf{u}^{(1)} \quad (15)$$

Note that $\mathbf{u}^{(1)}$ should include all global nodal displacements of the rod (i.e element in this case) 1 (solved nodal coordinates and nodal coordinates at supports). The member forces for rod 1 can be found as:

$$\bar{\mathbf{f}}^{(1)} = \bar{\mathbf{K}}^{(1)} \bar{\mathbf{u}}^{(1)} \quad (16)$$

and same operations for rods 2 and 3...